

Artificial Intelligence

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Lecture 7

Quantifying Uncertainty

- Prior probability
- Conditional probability
- Joint probability distribution
- Bayes' rule and its use
- Probabilistic reasoning (Bayesian network)
- Useful techniques for manipulating probability
- Naive Bayes Classifier

Uncertainty

In uncertain problems, the agent's knowledge can provide only a “degree of belief” in the sentences. Therefore, we need the probability theory to solve them.

Example: What is the probability that tomorrow it will rain in the morning and be sunny in the afternoon?

Prior probability

- Let $P(A)$ be the **unconditional** or **prior probability** that the proposition A is true.

Example: $P(\text{Cavity}) = 0.1$; for any patient.

- $P(A)$ can only be used when there is no other information.
- If some new information B is known, we have to reason with the **conditional probability** of A given B , $P(A | B)$, instead of $P(A)$.

Example of prior probability:

- Weather = { Hot, Rainy, Cold, Warm }
- $P(\text{Weather}) = \langle 0.7, 0.2, 0.08, 0.02 \rangle$
- $P(\text{Weather})$ defines a probability distribution for the random variable Weather.

Weather	$P(\text{Weather})$
Hot	0.7
Rainy	0.2
Cold	0.08
Warm	0.02

Conditional probability

- Once the agent has obtained more information, prior probabilities are no longer applicable.
- Instead, we use **conditional** or **posterior probability**, $P(A | B)$.

Example

$$P(\text{Cavity} | \text{Toothache}) = 0.8$$

↑
A Given B

- $P(A | B)$ can only be used when all we know is B.
- If we also know C, we must compute $P(A | B \wedge C)$ instead of $P(A | B)$.

Conditional Probability Distribution (One event occurs before another event)

$P(A|B)$

A

B		True	False
	Hot	0.4	0.6
	Cold	0.1	0.9

Note: $P(A|B) + P(\text{not } A|B) = 1$

- **Conditional probability** can be calculated by

$$P(A|B) = \frac{P(A \wedge B)}{P(B)} ; P(B) > 0.$$

- By moving a denominator to another side, we derive a **product rule** of two variables.

$$P(A \wedge B) = P(A|B) \cdot P(B)$$

- Note that $P(A \wedge B) \equiv P(A, B)$.

Joint Probability Distribution (Two events occur at the same time)

$P(\text{Toothache} \wedge \text{Cavity})$

	Toothache	$\neg$ Toothache
Cavity	0.04	0.06
$\neg$ Cavity	0.01	0.89

Sum to 1.0

Atomic event

- **Warning:** when you see a probability distribution table, it could be the conditional probability or the joint probability distribution. Read its description before using it.
- Atomic events are mutually exclusive.

$$P(\text{Cavity}) = 0.10$$

$$P(\text{Cavity} \vee \text{Toothache}) = \quad ?$$

$$P(\text{Cavity} \mid \text{Toothache}) = \quad ?$$

Bayes' rule and its use

$$P(A \wedge B) = P(A|B) \cdot P(B) \quad \dots\dots\dots(1)$$

$$P(A \wedge B) = P(B|A) \cdot P(A) \quad \dots\dots\dots(2)$$

$$(1) = (2) , \quad P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)} \quad ; \text{ Bayes' rule.}$$

$$\text{Similarly, } P(Y|X, E) = \frac{P(X|Y, E) \cdot P(Y|E)}{P(X|E)}$$

$$\begin{aligned} \text{Proof: } P(Y|X, E) &= \frac{P(Y, X, E)}{P(X, E)} \\ &= \frac{P(X, Y, E)}{P(X|E)P(E)} \\ &= \frac{P(X|Y, E)P(Y, E)}{P(X|E)P(E)} \\ &= \frac{P(X|Y, E)P(Y|E)P(E)}{P(X|E)P(E)} = \frac{P(X|Y, E)P(Y|E)}{P(X|E)} \end{aligned}$$

Example of Bayes' rule

- In medical studies, the doctor learns the symptoms of a given disease,
 $P(\text{symptoms} \mid \text{disease})$.
- In medical diagnosis, the doctor predicts a disease from the given symptoms.
 $P(\text{disease} \mid \text{symptoms})$.

For example, a doctor knows that the disease **meningitis** causes the patient to have a **stiff neck** 70% of the time.

$$P(s \mid m) = 0.7$$

The doctor also knows the prior probability that a patient has **meningitis** is 1/50,000,

$$P(m) = 1/50000$$

and the prior probability that any patient has a **stiff neck** is 1%.

$$P(s) = 0.01$$

We expect a patient with a **stiff neck** to have **meningitis** with probability

$$P(m \mid s) = \frac{P(s \mid m)P(m)}{P(s)} = \frac{0.7 \times 1/50000}{0.01} = 0.0014 .$$

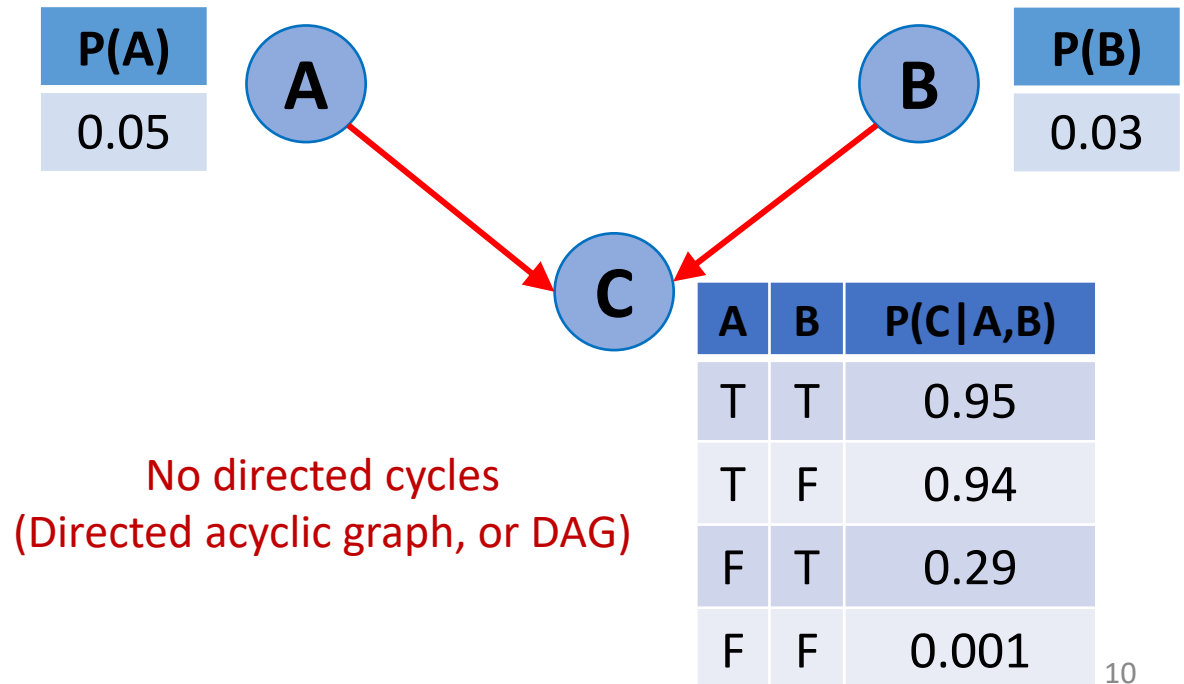
Probabilistic Reasoning Systems

We are asked to calculate the full joint probability distribution. If the number of variables grows, the matrices for keeping the probabilities become intractably large.

$$P(A \wedge B \wedge C \wedge D) = \overbrace{P(A|B \wedge C \wedge D).P(B \wedge C \wedge D)}^{\text{Not easy to calculate}}$$

Bayesian network

- A Bayesian network can reduce the joints of these variables.
- Graph represents the dependencies among variables.
- There is the joint or prior probability distribution for each variable.

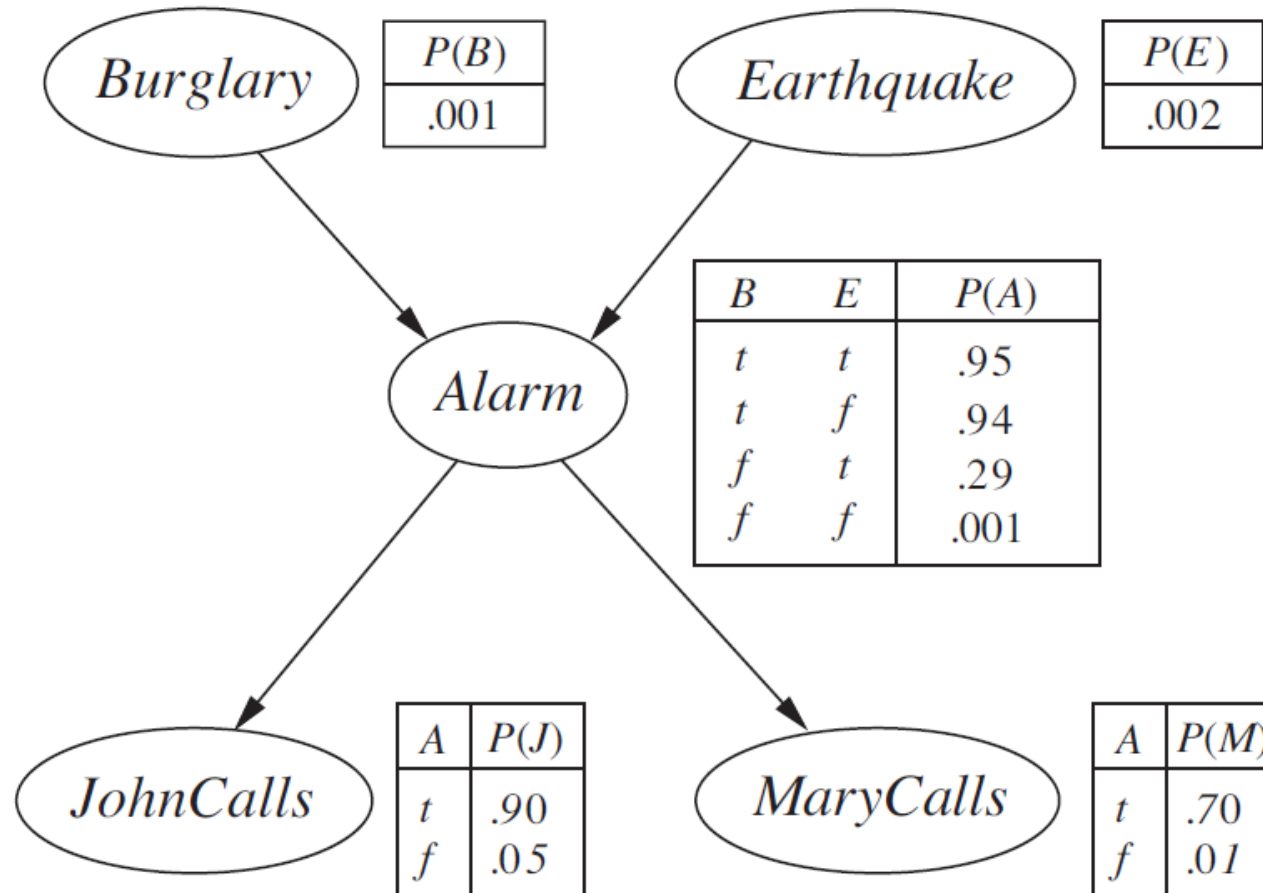


Example : Burglary alarm



- A new burglar alarm was installed at home. It is fairly reliable at detecting a burglary but also responds on occasion to minor earthquakes.
- You also have two neighbors, John and Mary, who have promised to call you at work when they hear the alarm.
- John sometimes confuses the telephone ringing with the alarm.
- Mary, on the other hand, likes loud music and often misses the alarm.
- **Query: What is the probability that the alarm has sounded, but neither a burglary nor an earthquake has occurred, and both John and Mary call?**

Drawing a directed acyclic graph, or DAG, to represent the Bayesian network.

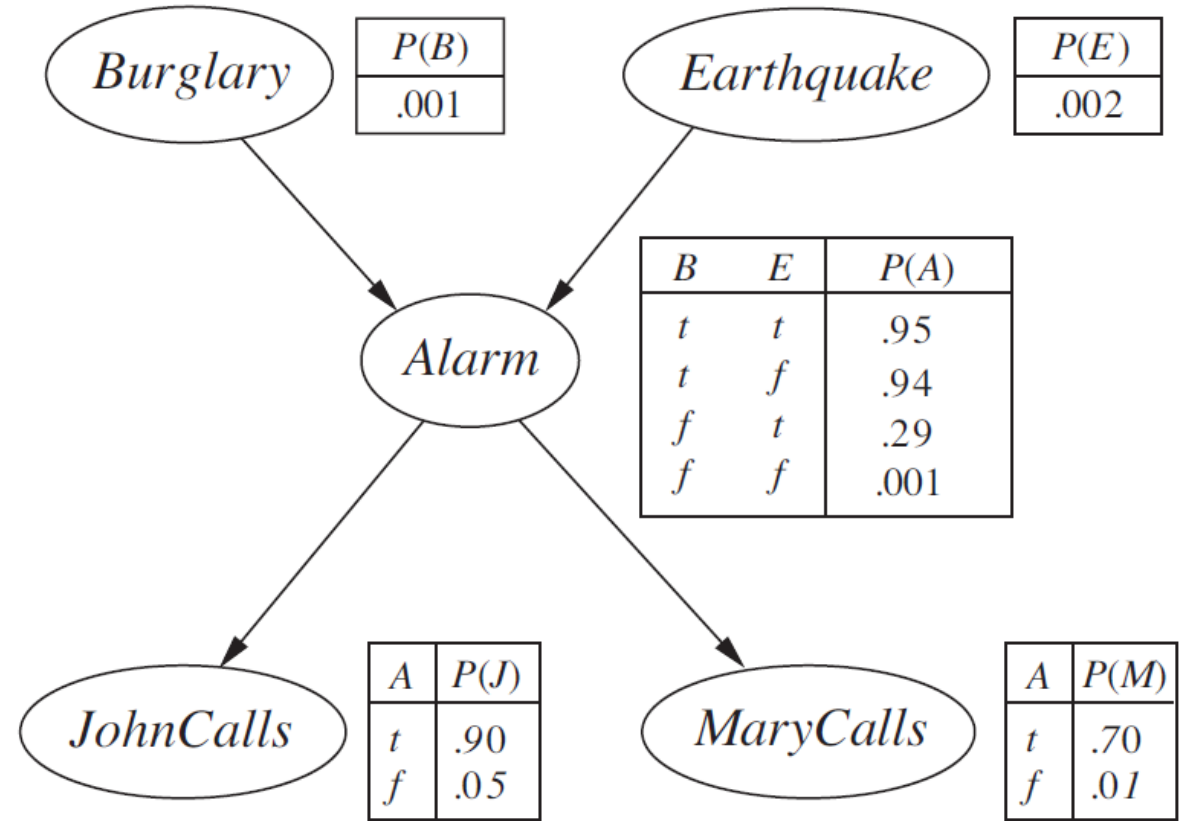


We can calculate the probability that **the alarm has sounded**, but **neither a burglary nor an earthquake has occurred**, and **both John and Mary call**.

$$P(j, m, a, \neg b, \neg e) = ?$$

The joint probability distribution can be represented by

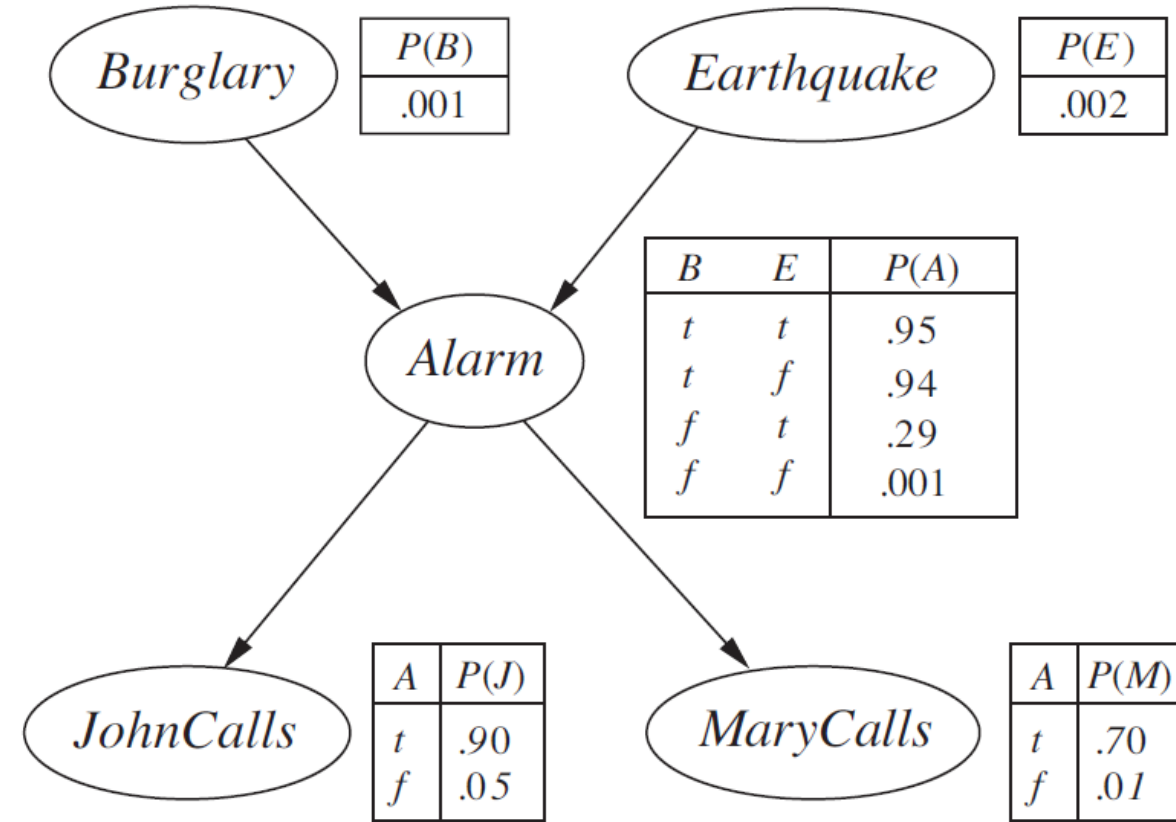
$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i \mid \text{parents}(X_i)) .$$



Note: Probability tables are filled by the experts.

We multiply entries from the joint distribution (using single-letter names for the variables):

b, e, a, j, and m stand for Burglary, Earthquake, Alarm, JohnCalls, and MaryCalls, respectively.



The joints of variables are significantly reduced.

$$\begin{aligned}
 P(j, m, a, \neg b, \neg e) &= \overbrace{P(j | a)P(m | a)} P(a | \neg b \wedge \neg e)P(\neg b)P(\neg e) \\
 &= 0.90 \times 0.70 \times 0.001 \times 0.999 \times 0.998 = 0.000628 .
 \end{aligned}$$

Useful techniques for manipulating probability

Marginalization

A domain consisting of 3 Boolean variables **Toothache**, **Cavity**, and **Catch** (whether the dentist's steel probe catches something wrong in my tooth).

$P(\text{Toothache}, \text{Cavity}, \text{Catch})$ is shown in 2x2x2 table as follows:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
$\neg$ <i>cavity</i>	0.016	0.064	0.144	0.576

Figure 13.3 A full joint distribution for the *Toothache*, *Cavity*, *Catch* world.

- Notice that the probabilities in the joint distribution sum to 1.
- $P(\text{cavity}) = 0.108 + 0.012 + 0.072 + 0.008 = 0.20$; **Marginal probability** of cavity
(or **summing out**)

- **Marginalization:** $P(Y) = \sum_{z \in Z} P(Y, z)$

Here we sum over all the possible values of the set of variables Z .

$$P(Cavity) = \sum_{z \in \{Catch, Toothache\}} P(Cavity, z)$$

- **Marginalization of Conditional Probability:**

Instead of using joint probabilities, we can apply the conditional probability to calculate the marginalization, $P(Y)$.

$$P(Y) = \sum_z P(Y|z)P(z)$$

- In most cases, we are more interested in computing the conditional probabilities rather than the full joint distribution since the size of probability table is smaller.

Normalization (Adjust the values measured on different scales to a common scale)

- Idea** 1st trial: Toss a coin 17 times, heads 10 times, tails 7 times. } This is hard to
2nd trial: Toss a coin 29 times, heads 15 times, tails 14 times. } interpret.
- $P_1(\text{head}) = 10/17 = 0.58$, $P_1(\text{tail}) = 7/17 = 0.42$ }
 $P_2(\text{head}) = 15/29 = 0.52$, $P_2(\text{tail}) = 14/29 = 0.48$ } Easier since their sum = 1.0

$$P(\text{cavity} \mid \text{toothache}) = \frac{P(\text{cavity} \wedge \text{toothache})}{P(\text{toothache})}$$
$$P(\neg \text{cavity} \mid \text{toothache}) = \frac{P(\neg \text{cavity} \wedge \text{toothache})}{P(\text{toothache})}$$

We can calculate $P(\text{Cavity} \mid \text{toothache})$ without knowing $P(\text{toothache})$!

- Let $1/P(\text{toothache})$ be a **normalization constant**, α .
- α can ensure that the distribution $P(\text{Cavity} \mid \text{toothache})$ adds up to 1.

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
$\neg$ <i>cavity</i>	0.016	0.064	0.144	0.576

Figure 13.3 A full joint distribution for the *Toothache*, *Cavity*, *Catch* world.

The Capital symbol **Cavity** means {cavity, $\neg$ cavity}.

$$\begin{aligned}
 \mathbf{P}(Cavity \mid toothache) &= \alpha \mathbf{P}(Cavity, toothache) \\
 &= \alpha [\mathbf{P}(Cavity, toothache, catch) + \mathbf{P}(Cavity, toothache, \neg catch)] \\
 &= \alpha [\langle 0.108, 0.016 \rangle + \langle 0.012, 0.064 \rangle] = \alpha \langle 0.12, 0.08 \rangle = \langle 0.6, 0.4 \rangle .
 \end{aligned}$$

$P(\text{cavity}, \text{toothache}, \text{catch})$

$P(\neg \text{cavity}, \text{toothache}, \text{catch})$

$P(\text{cavity}, \text{toothache}, \neg \text{catch})$

$P(\neg \text{cavity}, \text{toothache}, \neg \text{catch})$

Multiply each term by $\alpha = 1/(0.12+0.08) = 5$.

Marginalization

We can calculate $P(Cavity \mid toothache)$ without $P(toothache)$!

Independence between evidences in Bayes' rule

Bayes' rule can be viewed as the relationship between **cause** and **effect**.

$$\overbrace{P(\text{cause} | \text{effect})}^{\text{Diagnostic}} = \frac{\overbrace{P(\text{effect} | \text{cause})P(\text{cause})}^{\text{Causal}}}{P(\text{effect})}$$

What happens when we have two or more pieces of evidence?

$$\begin{aligned} \mathbf{P}(\text{Cavity} | \text{toothache} \wedge \text{catch}) \\ = \alpha \mathbf{P}(\text{toothache} \wedge \text{catch} | \text{Cavity}) \mathbf{P}(\text{Cavity}) \end{aligned}$$

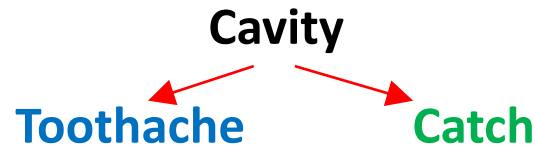
The joint of pieces of evidence makes it hard to compute.

Solution: The independence between variables can simplify the expression.

Using independence of variables to simplify Bayes' rule

In fact, **Toothache** and **Catch** are independent.

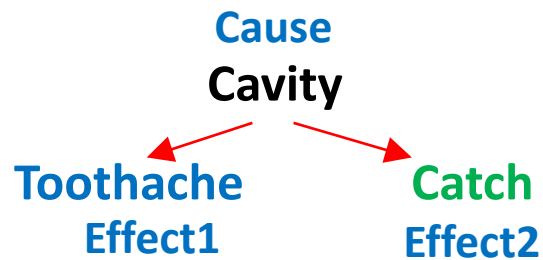
- **Toothache** depends on the nerves in the tooth
- **Catch** depends on the dentist's skill.



$$\begin{aligned} & \mathbf{P}(Cavity \mid toothache \wedge catch) \\ &= \alpha \mathbf{P}(toothache \wedge catch \mid Cavity) \mathbf{P}(Cavity) \\ &= \alpha \mathbf{P}(toothache \mid Cavity) \mathbf{P}(catch \mid Cavity) \mathbf{P}(Cavity) . \end{aligned}$$

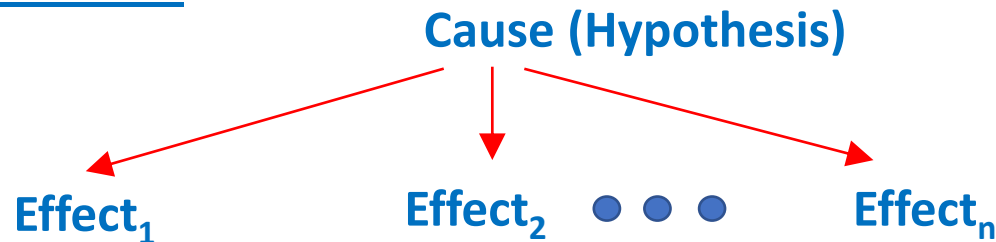
This is good because the joint probability disappears

We can also apply the **Independence of Evidence** to the Joint Probability Distribution.



$$\begin{aligned} & \mathbf{P}(\overset{\text{Effect1}}{Toothache}, \overset{\text{Effect2}}{Catch}, \overset{\text{Cause}}{Cavity}) \\ &= \mathbf{P}(\underline{Toothache}, \underline{Catch} \mid Cavity) \mathbf{P}(Cavity) \quad (\text{product rule}) \\ &= \mathbf{P}(\underline{Toothache} \mid Cavity) \mathbf{P}(\underline{Catch} \mid Cavity) \mathbf{P}(Cavity) \end{aligned}$$

General form



$$\mathbf{P}(Cause, Effect_1, \dots, Effect_n) = \mathbf{P}(Cause) \prod_i \mathbf{P}(Effect_i \mid Cause) .$$

- Such a probability distribution is called a **naive Bayes model**.
- The naive Bayes model is sometimes called a **Bayesian classifier**.

Naive Bayes Classifier

- Naive Bayes is a classifier that predicts the class based on the probability of the data points.
- The class with the highest probability is considered the most likely class, i.e., Maximum A Posteriori (MAP).
- The MAP for a hypothesis is:

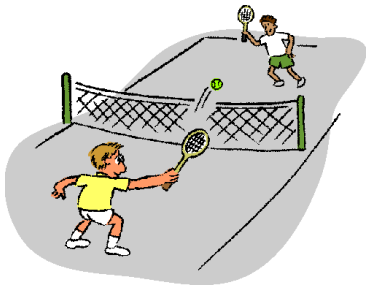
$$\begin{aligned} MAP(H) &= \max(P(H|E)) \\ &= \max\left(\frac{P(E|H)P(H)}{P(E)}\right) \\ &= \max(P(E|H) \cdot P(H)) \end{aligned}$$

Normalization constant
(removing it won't affect.)

- “Naïve” Bayes classifier assumes that all the evidences are unrelated to each other.

Simple example of Naive Bayes Classifier (single feature)

- Suppose we have a training data set of '**Weather**' and the target variable '**Play**' (tennis).
- We need to classify whether a player will **play or not** based on the **weather**.



14 instances

Weather	Play
Sunny	No
Overcast	Yes
Rainy	Yes
Sunny	Yes
Sunny	Yes
Overcast	Yes
Rainy	No
Rainy	No
Sunny	Yes
Rainy	Yes
Sunny	No
Overcast	Yes
Overcast	Yes
Rainy	No

Frequency Table		
Weather	No	Yes
Overcast		4
Rainy	3	2
Sunny	2	3
Grand Total	5	9

Convert a raw data into the frequency table.

Likelihood table				
Weather	No	Yes		
Overcast		4	=4/14	0.29
Rainy	3	2	=5/14	0.36
Sunny	2	3	=5/14	0.36
All	5	9		
	=5/14	=9/14		
	0.36	0.64		

Calculate the likelihood table.

- Will players play tennis if weather is **sunny**?

$$\begin{aligned} P(\text{Yes} \mid \text{Sunny}) &= P(\text{Sunny} \mid \text{Yes}) * P(\text{Yes}) / P(\text{Sunny}) \\ &= (3/9) * 0.64 / 0.36 = 0.60 \end{aligned}$$

$$\begin{aligned} P(\text{No} \mid \text{Sunny}) &= P(\text{Sunny} \mid \text{No}) * P(\text{No}) / P(\text{Sunny}) \\ &= (2/5) * 0.36 / 0.36 = 0.40 \end{aligned}$$

$\text{MAP}(H) = \max(0.60, 0.40) = 0.60$. Therefore, $H = \text{Yes}$.

Another example of Naive Bayes Classifier (multiple features)

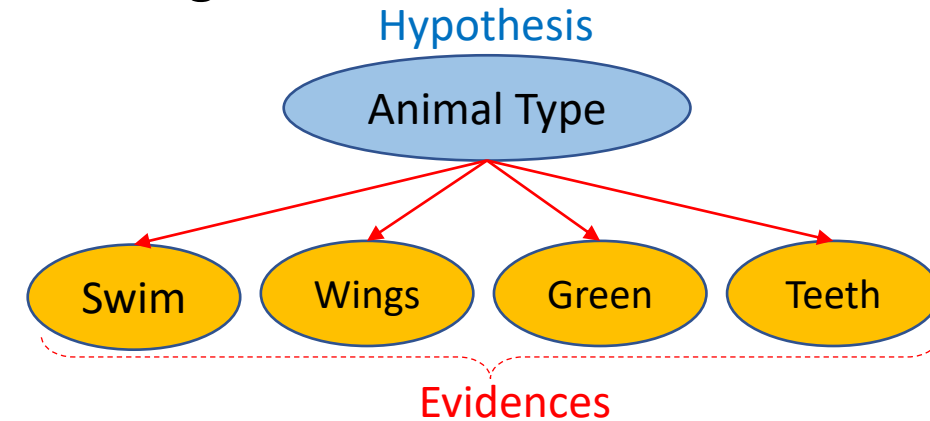


- We have 3 classes (500 instances each) associated with Animal Types:
Parrot, **Dog**, and **Fish**.
- There are 4 features (evidences) associated with each class:
Swim, **Wings**, **GreenColor**, and **Teeth**. => Categorical features T(True) or F(False)
- Frequency Table.

Swim	Wings	Green Color	Teeth	Animal Type
50/500	500/500	400/500	0	Parrot
450/500	0	0	500/500	Dog
500/500	0	100/500	50/500	Fish

- Predict whether the animal with the following features is a Dog, a Parrot, or a Fish.

Swim	Wings	Green Color	Teeth
True	False	True	False



- We will use the Naive Bayes approach

$$P(H|E1, E2, \dots, En) = P(E1|H) * P(E2|H) \dots * P(En|H) * P(H) / P(E1, E2, \dots, En)$$

- The Evidence here is Swim & Green. We ignore Teeth since it is False.
- $$P(\text{Dog} | \text{Swim, Green}) = P(\text{Swim} | \text{Dog}) * P(\text{Green} | \text{Dog}) * P(\text{Dog}) / P(\text{Swim, Green})$$

$$= 0.9 * 0 * 0.333 / P(\text{Swim, Green}) = \mathbf{0}$$

$$P(\text{Parrot} | \text{Swim, Green}) = P(\text{Swim} | \text{Parrot}) * P(\text{Green} | \text{Parrot}) * P(\text{Parrot}) / P(\text{Swim, Green})$$

$$= 0.1 * 0.80 * 0.333 / P(\text{Swim, Green}) = \mathbf{0.0264} / P(\text{Swim, Green})$$

$$P(\text{Fish} | \text{Swim, Green}) = P(\text{Swim} | \text{Fish}) * P(\text{Green} | \text{Fish}) * P(\text{Fish}) / P(\text{Swim, Green})$$

$$= 1 * 0.2 * 0.333 / P(\text{Swim, Green})$$

$$= \mathbf{0.0666} / P(\text{Swim, Green})$$

Therefore, the Naïve Bayes classifier predicts that this record is a fish.

Advantages

- Naive Bayes Algorithm is a simple, fast, scalable algorithm.
- It can be used for Binary and Multiclass classification.
- Great choice for text classification problems. It's a popular choice for spam email classification.

Disadvantages

- All of the evidences are assumed to be unrelated.
- It cannot learn if there is a relationship between evidences.

Ex: Remo likes to wear a **white shirt**. He likes to wear **brown jeans**.

But Remo doesn't wear a **white shirt** with **brown jeans**.